Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

The roots of cubic equation $(\mathrm{z}+\alpha\beta)^3=\alpha^3(\alpha\neq 0)$ represent the vertices of a triangle of sides of length -

(A) $\frac{1}{\sqrt{3}}|\alpha||\beta|$

(B) $\sqrt{3}|\alpha|$

(C) $\sqrt{3}|\beta|$

(D) $\frac{1}{\sqrt{3}}|\alpha|$

Q.2

If z is a complex number satisfying $|\mathrm{z}-\mathrm{i}\,\mathrm{Re}(\mathrm{z})|=|\mathrm{z}-\mathrm{Im}(\mathrm{z})|$ then z lies on $-$

(1) $\mathrm{y}=\mathrm{x}$

(2) $\mathrm{y}=-\mathrm{x}$

(3) $y=x+1$

(4) $y=-x+1$

(A) 1,2 and 3 are correct

(B) 1 and 2 are correct

(C) 2 and 4 are correct

(D) 1 and 3 are correct

Q.3

Let $\mathrm{C}=\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{8\pi}{7}$ and $S=\sin\frac{2\pi}{7}+\sin\frac{4\pi}{7}+\sin\frac{8\pi}{7}$, then

(A) $C=\frac{1}{2}$

(B) $S=\frac{1}{2}$

(C) $C=\frac{\sqrt{7}}{2}$

(D) $S=\frac{\sqrt{7}}{2}$

Q.4

The complex slope of a line passing through two points represented by complex numbers z_1 and z_2 is defined by $\frac{z_2-z_1}{\bar{z}_2-\bar{z}_1}$ and we shall denote by ω. If z_0 is complex number and c is a real number, then $\bar{z}_0z+z_0\bar{z}+c=0$ represents a straight line. Its complex slope is $\frac{-z_0}{\bar{z}_0}$.

Now consider two lines $\alpha\bar{z}+\bar{\alpha}z+i\beta=0$ ...(i)

and $a\bar{z}+\bar{a}z+b=0$ ...(ii)

@IITJEE_Advanced

where α, β and a, b are complex constants and let their complex slopes be denoted by ω_1 and ω_2 respectively - If the lines are inclined at an angle of 120° to each other, then -

(A) $\omega_2\bar{\omega}_1 = \omega_1\bar{\omega}_1$

(B) $\omega_2\bar{\omega}_1^2 = \omega_1\bar{\omega}_2^2$

(C) $\omega_1^2 = \omega_2^2$

(D) $\omega_1 + 2\omega_2 = 0$

Q.5

The complex slope of a line passing through two points represented by complex numbers z_1 and z_2 is defined by $\frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$ and we shall denote by ω. If z_0 is complex number and c is a real number, then $\bar{z}_0 z + z_0\bar{z} + c = 0$ represents a straight line. Its complex slope is $\frac{-z_0}{\bar{z}_0}$.
Now consider two lines $\alpha\bar{z} + \bar{\alpha}z + i\beta = 0$ ...(i)
and $a\bar{z} + \bar{a}z + b = 0$ ...(ii)
where α, β and a, b are complex constants and let their complex slopes be denoted by ω_1 and ω_2 respectively - Which of the following must be true -

(A) a must be pure imaginary

(B) β must be pure imaginary

(C) a must be real

(D) b must be imaginary

Q.6

The complex slope of a line passing through two points represented by complex numbers z_1 and z_2 is defined by $\frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$ and we shall denote by ω. If z_0 is complex number and c is a real number, then $\bar{z}_0 z + z_0\bar{z} + c = 0$ represents a straight line. Its complex slope is $\frac{-z_0}{\bar{z}_0}$.
Now consider two lines $\alpha\bar{z} + \bar{\alpha}z + i\beta = 0$ ...(i)
and $a\bar{z} + \bar{a}z + b = 0$ ...(ii)
where α, β and a, b are complex constants and let their complex slopes be denoted by ω_1 and ω_2 respectively - If line (i) makes an angle of 45° with real axis, then $(1+i)\left(-\frac{2\alpha}{\bar{\alpha}}\right)$ is $-$

(A) $2\sqrt{2}$

(B) $2\sqrt{2}i$

(C) $2(1 - \mathrm{i})$

(D) $-2(1+i)$

Q.7

If a, b, c are integers not all equal and ω is a complex cube root of unity, then the minimum value of $\left|a + b\omega + c\omega^2\right|$ is

(A) 0

(B) 1

@IITJEE_Advanced

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2}$

Q.8

If α and β are roots of the equation $x^2 - 2x + 4 = 0$ then find the value of $\alpha^{12} + \beta^{12}$

(A) 7^{13}

(B) 3^{13}

(C) 5^{13}

(D) 2^{13}

Q.9

If z_1, z_2 be two complex numbers $(z_1 \neq z_2)$ satisfying $\left|z_1^2 - z_2^2\right| = \left|\bar{z}_1^2 + \bar{z}_2^2 - 2\bar{z}_1\bar{z}_2\right|$, then $-$

(A) $\frac{z_1}{z_2}$ is purely imaginary

(B) $\frac{z_1}{z_2}$ is purely real

(C) $|\arg z_1 - \arg z_2| = \pi$

(D) $|\arg z_1 - \arg z_2| = \pi/2$

Q.10

If $\left|\frac{z-z_1}{z-z_2}\right| = 3$, where z_1 and z_2 are fixed complex numbers and z is a variable complex number, then z lies on $a-$

(A) Circle with z_1 as its interior point

(B) Circle with z_2 as its interior point

(C) Circle with z_1 as its exterior point

(D) Circle with z_2 as its exterior point

Q.11

Locus of the point z satisfying the equation $|z - z_1| + |z - z_2| = \lambda, \lambda \in R^+$ and $\lambda \not< |z_1 - z_2|$

(A) A straight line

(B) A parabola

(C) An ellipse

(D) A circle

Q.12

Passage:
Consider two complex number z_1 and z_2 satisfying $|z| = 1$ and $|z - 1| = 1$ respectively.

@IITJEE_Advanced

On the basis of above information answer the following :

Question:

Minimum value of $|z_1 - z_2|$ is equal to -

(A) 0

(B) 1

(C) 2

(D) None of these

Q.13

Passage:

Consider two complex number z_1 and z_2 satisfying $|z| = 1$ and $|z - 1| = 1$ respectively.

On the basis of above information answer the following :

Question:

Maximum value of $|z_1 - z_2|$ is equal to $-$

(A) 1

(B) 2

(C) 3

(D) 4

Q.14

Passage:

Consider two complex number z_1 and z_2 satisfying $|z| = 1$ and $|z - 1| = 1$ respectively.

On the basis of above information answer the following :

Question:

If the range of amp $z_1 \in (-a, a]$ and that of amp $z_2 \in (-b, b)$, then $\frac{a}{b}$ is equal to -

(A) 1

(B) 2

(C) $1/2$

(D) does not exist

Q.15

Passage:

Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers.

Question:

The condition that the equation has one purely imaginary root is -

(A) $(c\bar{a} - a\bar{c})^2 = -(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$

(B) $(c\bar{a} + a\bar{c})^2 = (b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$

(C) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b)$

@IITJEE_Advanced

(D) None of these

Q.16

Passage:
Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers.
Question:
If equation has two purely imaginary roots, then which of the following is not true -

(A) $a\bar{b}$ is purely imaginary

(B) $b\bar{c}$ is purely imaginary

(C) $c\bar{a}$ is purely real

(D) None of these

Q.17

Passage:
Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers.
Question:
The condition that the equation has one purely real root is -

(A) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} + c\bar{b})(a\bar{b} - \bar{a}b)$

(B) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} + \bar{a}b)$

(C) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$

(D) $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b)$

Q.18

Match the column (Quadrilateral is formed by z_1, z_2, z_3, z_4 in the order)

Column I	**Column II**
(A) Parallelogram	(p) $z_1 - z_4 = z_2 - z_3$
(B) Rectangle	(q) $\lvert z_1 - z_3 \rvert = \lvert z_2 - z_4 \rvert$
(C) Rhombus	(r) $\frac{z_1 - z_2}{z_3 - z_4}$ is purely real
(D) Square	(s) $\frac{z_1 - z_3}{z_2 - z_4}$ is purely imaginary
	(t) $\frac{z_1 - z_2}{z_3 - z_2}$ is purely imaginary

(A) (A) - p,q, (B) - q,r, (C) - p,q,s, (D) - s,t

(B) (A) - p,r, (B) - p,q,r,t, (C) - p,r,s, (D) - p,q,r,s,t

(C) (A) - r,s, (B) - s,r,t, (C) - p,q,s,t, (D) - r,s

@IITJEE_Advanced

(D) (A) - p,q,r,s, (B) - r,s,t, (C) - p,q,t, (D) - q,r,s

Q.19

The inequality $|z-4|<|z-2|$ represents the region given by $\operatorname{Re}(z)>A$. Find the value of A.

Q.20

If $\frac{1}{x}+x=2\cos\theta, x^n+\frac{1}{x^n}=A\cos n\theta$. Find the value of A.

Answers & Solutions

Q.1 Answer:

$\sqrt{3}|\alpha|$

Solution:

$$\because \left(\frac{z+\alpha\beta}{\alpha}\right)^3=1 \quad \therefore \frac{z+\alpha\beta}{\alpha}=1,\omega,\omega^2$$

$\therefore \quad z+\alpha\beta=\alpha,\omega\alpha,\omega^2\alpha$

$\therefore$ vertices are $A(\alpha-\alpha\beta), B(\omega\alpha-\alpha\beta), C\left(\omega^2\alpha-\alpha\beta\right)$

$\therefore$ Length of $AB=|z_1-z_2|=|(\alpha-\alpha\beta)-(\omega\alpha-\alpha\beta)|$

$=|\alpha-\omega\alpha|=|1-\omega||\alpha|=\sqrt{3}|\alpha|$

Q.2 Answer:

1 and 2 are correct

Solution:

$|z-i\operatorname{Re}(z)|=|z-\operatorname{Im}(z)|$

Let $z=x+iy$**, then** $|x+iy-ix|=|x+iy-y|$

i.e. $x^2+(y-x)^2=(x-y)^2+y^2$

i.e. $x^2=y^2$ **i.e.** $y=\pm x$

Q.3 Answer:

$S=\frac{\sqrt{7}}{2}$

Solution:

$C+iS=e^{i\theta}+e^{i(2\theta)}+e^{i(4\theta)}$**, where** $\theta=\frac{2\pi}{7}$

i .e $C+iS=\alpha+\alpha^2+\alpha^4$**, where** $\alpha=e^{i\theta}$ **...(i)**

so, $C-iS=\bar{\alpha}+\left(\bar{\alpha}^2\right)+\left(\bar{\alpha}^4\right)=\alpha^6+\alpha^5+\alpha^3$ **...(ii)**

$\left(\bar{\alpha}=\frac{\alpha\bar{\alpha}}{\alpha}=\frac{1}{\alpha}=\frac{\alpha^7}{\alpha}=\alpha^6\right.$ **etc** $)$

From Eqs (i) and (ii), we get

$2C=\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5+\alpha^6=\frac{\alpha\left(\alpha^6-1\right)}{\alpha-1}$

$\Rightarrow 2C=\frac{1-\alpha}{\alpha-1}\Rightarrow C=-\frac{1}{2}$

@IITJEE_Advanced

Again $(C+iS)(C-iS)=\left(\alpha+\alpha^2+\alpha^4\right)\left(\alpha^6+\alpha^5+\alpha^3\right)$
$\Rightarrow C^2+S^2=1+\alpha^6+\alpha^4+\alpha+1+\alpha^5+\alpha^3+\alpha^2+1\left(a^7=1\right)$
$\Rightarrow\left(-\frac{1}{2}\right)^2+S^2=2+\left(1+\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5+\alpha^6\right)$
$\Rightarrow S^2=2+0-\frac{1}{4}=\frac{7}{4}$
$\Rightarrow S=\frac{\sqrt{7}}{2}$

Q.4 Answer:

$\omega_2\bar{\omega}_1^2=\omega_1\bar{\omega}_2^2$

Solution:

$\omega_1=\omega_2 e^{i\frac{4\pi}{3}} \Rightarrow \omega_1^3=\omega_2^3 \Rightarrow \omega_1^3\bar{\omega}_1^2\bar{\omega}_2^2=\omega_2^3\bar{\omega}_1^2\bar{\omega}_2^2 \Rightarrow \omega_2\bar{\omega}_1^2=\omega_1\bar{\omega}_2^2$

Q.5 Answer:

β must be pure imaginary

Solution:

Since i β **is real**
$\therefore$ β **pure imaginary**

Q.6 Answer:

$2(1-i)$

Solution:

$-\frac{\alpha}{\bar{\alpha}}=e^{\pm\frac{i\pi}{2}}=\pm i \quad \therefore(1+i)\left(-\frac{2\alpha}{\bar{\alpha}}\right)=\pm 2(-1+i)$

Q.7 Answer:

1

Solution:

$$\left|a+b\omega+c\omega^2\right|^2=\left(a+b\omega+c\omega^2\right)\overline{\left(a+b\omega+c\omega^2\right)}$$
$$=\left(a+b\omega+c\omega^2\right)\left(a+b\omega^2+c\omega\right)$$
$$=a^2+b^2+c^2+ab\left(\omega+\omega^2\right)+bc\left(\omega^2+\omega^4\right)+ca\left(\omega+\omega^2\right)$$
$$=a^2+b^2+c^2-ab-bc-ca=\frac{1}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$
has minimum value 1 for $a=b=1, c=2$

Q.8 Answer:

2^{13}

Solution:

Solving the equation $x^2-2x+4=0$
we get $\alpha=1+i\sqrt{3}; \beta=1-i\sqrt{3}$
Here $\alpha^{12}+\beta^{12}=(1+i\sqrt{3})^{12}+(1-i\sqrt{3})^{12}$

@IITJEE_Advanced

Now $1+i\sqrt{3}=2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$ and $\left(1-i\sqrt{3}\right)=2\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)$

$$\therefore \quad \alpha^{12}+\beta^{12}=\left[2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\right]^{12}+\left[2\left(\cos\left(\frac{\pi}{3}\right)-i\sin\left(\frac{\pi}{3}\right)\right)\right]^{12}$$

$$=2^{12}[\cos 4\pi+i\sin 4\pi]+2^{12}[\cos 4\pi-i\sin 4\pi]=2^{12}(1+0)+2^{12}(1-0)$$

$$=2^{12}+2^{12}=2^{12}(1+1)$$

$$\therefore \quad \alpha^{12}+\beta^{12}=2.2^{12}=2^{13}$$

Q.9 Answer:

$\frac{z_1}{z_2}$ is purely imaginary

Solution:

$|z_1+z_2|=|z_1-z_2| \Rightarrow \left|\frac{z_1}{z_2}+1\right|=\left|\frac{z_1}{z_2}-1\right| \Rightarrow$ lies on $\perp$ bisector of 1 and $-1 \Rightarrow$ $\frac{z_1}{z_2}$ lies on imaginary axis $\Rightarrow$ $\frac{z_1}{z_2}$ is purely imaginary

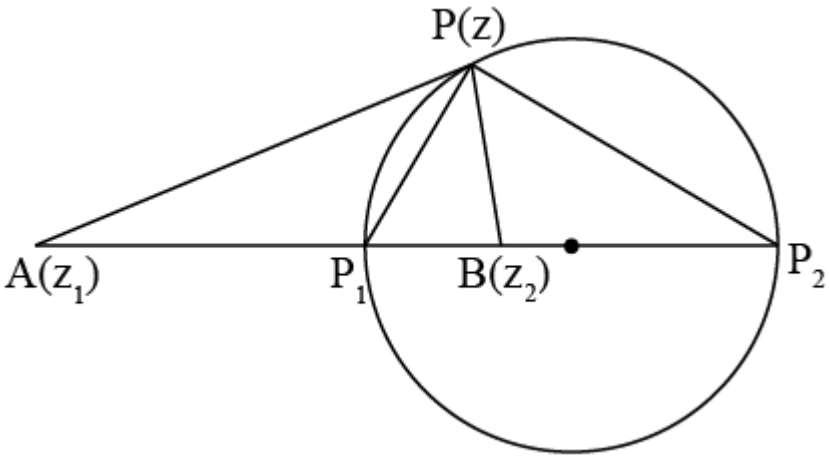
$$\Rightarrow \quad \arg\left(\frac{z_1}{z_2}\right)=\pm\frac{\pi}{2}$$

$$|\arg(z_1)-\arg(z_2)|=\frac{\pi}{2}$$

Q.10 Answer:

Circle with z_2 as its interior point

Solution:



Let internal and external bisectors of $\angle$ APB meet the line joining A and B at P_1 and P_2, respectively.

Hence, $AP_1 : P_1B \equiv PA : PB \equiv 3:1$ (internal division) $AP_2 : P_2B \equiv PA : PB \equiv 3:1$ (external division)

Thus, P_1 and P_2 are fixed points.

Also,

$\angle P_1PP_2=\frac{\pi}{2}$

Thus, P lies on a circle having P_1P_2 as its diameter.

Clearly, $B\,(z_2)$ lies inside this circle.

Q.11 Answer:

A straight line

@IITJEE_Advanced

Solution:

For ellipse $\lambda > |z_1 - z_2|$ and for straight line $\lambda = |z_1 - z_2|$

Q.12 Answer:

0

Solution:



z_1 lies on a circle with centre $(0, 0)$ and radius 1.

z_2 also lies on a circle with centre $(1, 0)$ and radius 1.

(i) Minimum value of $|z_1 - z_2| = 0$ because both the circles are intersecting.

(ii) Maximum value of $|z_1 - z_2| = A'B = 3$

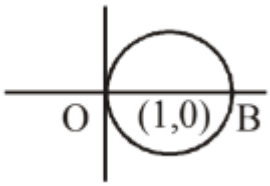


(iii) $\text{amp}\, z_1 \in (-a, a]$ according to figure.

$-\pi < \text{amp}\, z_1 \leq \pi$

hence $a = \pi$

amp. $z_2 \in (-b, b)$



According to figure, $-\frac{\pi}{2} < \text{amp} z_2 < \frac{\pi}{2}$

hence $b = \frac{\pi}{2}$; $\frac{a}{b} = 2$

Q.13 Answer:

3

Solution:



@IITJEE_Advanced

z_1 lies on a circle with centre $(0, 0)$ and radius 1.

z_2 also lies on a circle with centre $(1, 0)$ and radius 1.

(i) Minimum value of $|z_1 - z_2| = 0$ because both the circles are intersecting.

(ii) Maximum value of $|z_1 - z_2| = A'B = 3$

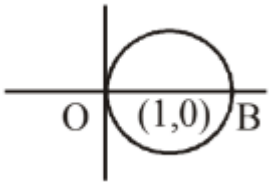


(iii) $\text{amp}\, z_1 \in (-a, a]$ according to figure.

$-\pi < \text{amp}\, z_1 \leq \pi$

hence $a = \pi$

amp. $z_2 \in (-b, b)$



According to figure, $-\frac{\pi}{2} < \text{amp} z_2 < \frac{\pi}{2}$

hence $b = \frac{\pi}{2}$; $\frac{a}{b} = 2$

Q.14 Answer:

2

Solution:



z_1 lies on a circle with centre $(0, 0)$ and radius 1.

z_2 also lies on a circle with centre $(1, 0)$ and radius 1.

(i) Minimum value of $|z_1 - z_2| = 0$ because both the circles are intersecting.

(ii) Maximum value of $|z_1 - z_2| = A'B = 3$



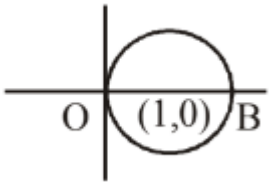
(iii) $\text{amp}\, z_1 \in (-a, a]$ according to figure.

$-\pi < \text{amp}\, z_1 \leq \pi$

@IITJEE_Advanced

hence $a = \pi$

amp. $z_2 \in (-b, b)$



According to figure, $-\frac{\pi}{2} < \text{amp}z_2 < \frac{\pi}{2}$

hence $b = \frac{\pi}{2}; \frac{a}{b} = 2$

Q.15 Answer:

$(c\bar{a} - a\bar{c})^2 = -(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b)$

Solution:

(i) Let z_1 **(purely imaginary) be a root of the given equation.**

Then, $z_1 = -\bar{z}_1$ **and** $az_1^2 + bz_1 + c = 0$ **....(1)**

$\Rightarrow \quad \bar{a}\bar{z}_1^2 + \bar{b}\bar{z}_1 + \bar{c} = 0$

$\Rightarrow \quad \bar{a}z_1^2 - \bar{b}z_1 + \bar{c} = 0 \quad (\text{as } \bar{z}_1 = -z_1) \ldots.(2)$

Now, eq. (1) and (2) must have one common root.

$\therefore \quad (c\bar{a} - a\bar{c})^2 = (b\bar{c} + c\bar{b})(-a\bar{b} - \overline{ab})$

(ii) Let z_1 **and** z_2 **be two purely imaginary roots.**

Then, $\bar{z}_1 = -z_1, \bar{z}_2 = -z_2$

Now, $az^2 + bz + c = 0$

$\Rightarrow \quad \overline{az^2 + bz + c} = \bar{0}$ **...(1)**

$\Rightarrow \quad \bar{a}\bar{z}^2 + \bar{b}\bar{z} + \bar{c} = 0$

$\Rightarrow \quad \bar{a}z^2 - \bar{b}z + \bar{c} = 0$ **...(2)**

Equations (1) and (2) must be identical as their roots are same.

$\therefore \quad \frac{\bar{a}}{a} = -\frac{\bar{b}}{b} = \frac{\bar{c}}{c}$

$\Rightarrow \quad a\bar{c} = \bar{a}c, a\bar{b} + \bar{a}b = 0 \text{ and } b\bar{c} + \bar{b}c = 0$

Hence, $a\bar{c}$ **is purely real and** $a\bar{b}$ **and** $b\bar{c}$ **are purely imaginary.**

(iii) Let z_1 **(purely real) be a root of the given equation.**

Then $z_1 = \bar{z}_1$ **and** $az_1^2 + bz_1 + c = 0$ **...(1)**

$\Rightarrow \quad \overline{az_1^2 + bz_1 + c} = \bar{0}$

$\Rightarrow \quad \overline{a}\overline{z}_1^2 + \bar{b}\bar{z}_1 + \bar{c} = 0$

$\Rightarrow \quad \bar{a}z_1^2 + \bar{b}z_1 + \bar{c} = 0 \ldots(2)$

Now, eq. (1) and (2) must have one common root.

Hence, $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b)$

Q.16 Answer:

None of these

Solution:

(i) Let z_1 **(purely imaginary) be a root of the given equation.**

Then, $z_1 = -\bar{z}_1$ **and** $az_1^2 + bz_1 + c = 0$ **....(1)**

@IITJEE_Advanced

$\Rightarrow \quad \overline{a}\overline{z}_1^2 + \overline{b}\overline{z}_1 + \overline{c} = 0$

$\Rightarrow \quad \overline{a}z_1^2 - \overline{b}z_1 + \overline{c} = 0 \quad (\text{as } \overline{z}_1 = -z_1) \ldots.(2)$

Now, eq. (1) and (2) must have one common root.

$\therefore \quad (c\bar{a} - a\bar{c})^2 = (b\bar{c} + c\bar{b})(-a\bar{b} - \overline{ab})$

(ii) Let z_1 and z_2 be two purely imaginary roots.

Then, $\overline{z}_1 = -z_1, \overline{z}_2 = -z_2$

Now, $az^2 + bz + c = 0$

$\Rightarrow \quad \overline{az^2 + bz + c} = \overline{0}$ **...(1)**

$\Rightarrow \quad \overline{a}\overline{z}^2 + \overline{b}\overline{z} + \overline{c} = 0$

$\Rightarrow \quad \overline{a}z^2 - \overline{b}z + \overline{c} = 0$ **...(2)**

Equations (1) and (2) must be identical as their roots are same.

$\therefore \quad \frac{\overline{a}}{a} = -\frac{\overline{b}}{b} = \frac{\overline{c}}{c}$

$\Rightarrow \quad a\overline{c} = \overline{a}c, a\overline{b} + \overline{a}b = 0 \text{ and } b\overline{c} + \overline{b}c = 0$

Hence, $a\bar{c}$ is purely real and $a\bar{b}$ and $b\bar{c}$ are purely imaginary.

(iii) Let z_1 (purely real) be a root of the given equation. Then $z_1 = \overline{z}_1$ and $az_1^2 + bz_1 + c = 0$ ...(1)

$\Rightarrow \quad \overline{az_1^2 + bz_1 + c} = \overline{0}$

$\Rightarrow \quad \overline{a}\overline{z}_1^2 + \overline{b}\overline{z}_1 + \overline{c} = 0$

$\Rightarrow \quad \bar{a}z_1^2 + \bar{b}z_1 + \bar{c} = 0 \ldots(2)$

Now, eq. (1) and (2) must have one common root.

Hence, $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \overline{a}b)$

Q.17 Answer:

$(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \overline{a}b)$

Solution:

(i) Let z_1 (purely imaginary) be a root of the given equation.

Then, $z_1 = -\overline{z}_1$ **and** $az_1^2 + bz_1 + c = 0$ **....(1)**

$\Rightarrow \quad \overline{a}\overline{z}_1^2 + \overline{b}\overline{z}_1 + \overline{c} = 0$

$\Rightarrow \quad \overline{a}z_1^2 - \overline{b}z_1 + \overline{c} = 0 \quad (\text{as } \overline{z}_1 = -z_1) \ldots.(2)$

Now, eq. (1) and (2) must have one common root.

$\therefore \quad (c\bar{a} - a\bar{c})^2 = (b\bar{c} + c\bar{b})(-a\bar{b} - \overline{ab})$

(ii) Let z_1 and z_2 be two purely imaginary roots.

Then, $\overline{z}_1 = -z_1, \overline{z}_2 = -z_2$

Now, $az^2 + bz + c = 0$

$\Rightarrow \quad \overline{az^2 + bz + c} = \overline{0}$ **...(1)**

$\Rightarrow \quad \overline{a}\overline{z}^2 + \overline{b}\overline{z} + \overline{c} = 0$

$\Rightarrow \quad \overline{a}z^2 - \overline{b}z + \overline{c} = 0$ **...(2)**

Equations (1) and (2) must be identical as their roots are same.

$\therefore \quad \frac{\overline{a}}{a} = -\frac{\overline{b}}{b} = \frac{\overline{c}}{c}$

$\Rightarrow \quad a\overline{c} = \overline{a}c, a\overline{b} + \overline{a}b = 0 \text{ and } b\overline{c} + \overline{b}c = 0$

Hence, $a\bar{c}$ is purely real and $a\bar{b}$ and $b\bar{c}$ are purely imaginary.

(iii) Let z_1 (purely real) be a root of the given equation.

Then $z_1 = \overline{z}_1$ and $az_1^2 + bz_1 + c = 0$ ...(1)

$\Rightarrow \quad \overline{az_1^2 + bz_1 + c} = \overline{0}$

@IITJEE_Advanced

$\Rightarrow \quad \overline{a}\overline{z}_1^2 + \overline{b}\overline{z}_1 + \overline{c} = 0$

$\Rightarrow \quad \bar{a}z_1^2 + \bar{b}z_1 + \bar{c} = 0 \ldots (2)$

Now, eq. (1) and (2) must have one common root.

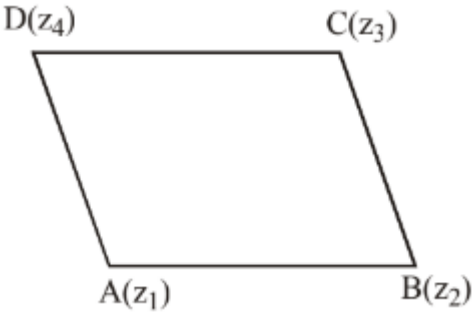
Hence, $(c\bar{a} - a\bar{c})^2 = (b\bar{c} - c\bar{b})(a\bar{b} - \bar{a}b)$

Q.18 Answer:

(A) - p,r, (B) - p,q,r,t, (C) - p,r,s, (D) - p,q,r,s,t

Solution:

(A) $-\mathbf{pr}, (B) - \mathbf{pqrt}, (C) - \mathbf{prs}, (D) - \mathbf{pqrst}$



In parallelogram, the mid-points of diagonals coincide

$\frac{z_1+z_3}{2} = \frac{z_2+z_4}{2} \Rightarrow z_1 - z_4 = z_2 - z_3$

Also in parallelogram, AB $\|$ **CD.**

Hence, $\arg\left(\frac{z_1-z_2}{z_3-z_4}\right) = 0 \Rightarrow \frac{z_1-z_2}{z_3-z_4}$ **is purely real**

In rectangle, adjacent sides are perpendicular.

Hence, $\arg\left(\frac{z_1-z_3}{z_3-z_2}\right) = \frac{\pi}{2} \Rightarrow \frac{z_1-z_2}{z_3-z_2}$ **is purely imaginary.**

Also in rectangle, $AB = BD \Rightarrow |z_1 - z_3| = |z_2 - z_4|$

In rhombus, $AC \perp BD \Rightarrow \frac{z_1-z_3}{z_2-z_4}$ **is purely imaginary**

Q.19 Answer:

—

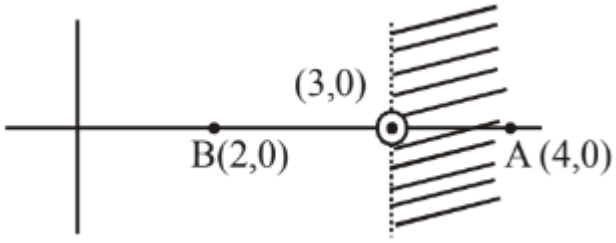
Solution:

Let $P(z), A(4,0), B(2,0)$

$\therefore \quad |z-4| < |z-2|$

$\Rightarrow \quad PA < PB$

$\therefore \quad \mathrm{Re}(z) > 3$



Q.20 Answer:

—

@IITJEE_Advanced

Solution:

$\frac{1}{x}+x=2\cos\theta \quad \Rightarrow \quad x^2-2x\cos\theta+1=0$
$\Rightarrow \quad x=\cos\theta\pm i\sin\theta \quad \Rightarrow \quad x^n=\cos n\theta\pm i\sin n\theta$
$\Rightarrow \frac{1}{x}=\frac{1}{\cos\theta\pm i\sin\theta} \quad \Rightarrow \quad \frac{1}{x}=\cos\theta\overline{+}i\sin\theta$
$\Rightarrow \quad \frac{1}{x^n}=\cos n\theta\mp i\sin n\theta \quad$ **Thus,** $x^n+\frac{1}{x^n}=2\cos n\theta$

@IITJEE_Advanced